Viscar Industrial Capacity Limited

Enhancing Performance & Productivity

SUPERVISORY CONTROL and DATA ACQUISITION (SCADA) SYSTEMS MANAGEMENT COURSE OUTLINE

TRAINING, CONSULTANCY & RESOURCING

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SUPERVISORY CONTROL and DATA ACQUISITION (SCADA) SYSTEMS MANAGEMENT

Duration: 5 Days

Course Overview

SCADA has traditionally meant a window into the process of a plant and/or a method of gathering of Data from devices in the field. Today, the focus is on integrating this process data into the actual business, and using it in real time. In addition to this, today‟s emphasis is on using Open Standards, such as communication protocols (e.g. IEC 104, DNP3 and TCP/IP) and „off-the-shelf‟ hardware and software, as well as focusing on keeping the costs down. PLCs continue to gain in popularity. This comprehensive workshop covers the essentials of SCADA and PLC systems which are often used in close association with each other. A selection of a case studies are used to illustrate the key concepts with examples of real world working SCADA and PLC systems in the electrical and Processing industries. This training will be an excellent opportunity to significant new information and Techniques for your next SCADA/PLC project

The training will also put emphasis on the current SCADA trends and drivers as well as the latest technologies available.

The Course is built around:

i. Understanding of a SCADA System – Types of SCADA Systems
ii. SCADA Software
iii. SCADA Graphic – Creating and assigning SCADA Graphics
iv. Launching runtime
v. Simulation of actual industrial process flow problems

Target Audience

- Plant engineers
- Electrical maintenance technicians and supervisors
- Instrument and control engineers and technicians
- Process control engineers
- Mechanical personnel

COURSE OUTLINE

Background to SCADA

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- Fundamentals and definitions of terms
- Comparison of SCADA, DCS, PLC and Smart instruments
- Typical SCADA installations
- Trends- Real Time & Historical

SCADA System Hardware

- Comparison of SCADA, DCS, PLC and smart instruments
- Remote Terminal Unit(RTU) Structure
- Analogue and Digital input/output modules
- PLCs Used as RTUs
- Master site structure
- Communications architectures
- Point –to-point and point-to-multi point systems
- Configurations of a master station

SCADA Systems Software

- Components of a SCADA system
- Software –design of SCADA packages
- Configuration of SCADA systems
- Building the user interface
- Connecting to PLCs and other hardware
- SCADA system design
- The twelve golden rules

SCADA Alarm Management

- Alarm layout and organization
- Alarm priorities
- Alarm processing and reporting

Creating Database / New SCADA Project

- Database & DDE connectivity

The use of Industrial Ethernet in SCADA systems
Communication interfaces

- ERP systems such as SAP R/3, LIMS systems, maintenance systems

Communication protocols in SCADA

Troubleshooting SCADA Systems

- Problem isolation
- Testing methodology
- Noise issues
- Communications testing

Human Management Interface (HMI)

- Ergonomic factors

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- HMI organization
- HMI screen design

Good Programming Habits

- Keeping track of addresses and data used
- Looking ahead-how will programs be maintained?
- Practical methods to improve quality, organization of code, documentation and simplifying changes

Good Installation Practice

- Location of hardware
- Good wiring practice
- Cable spacing, power distribution and wire numbering
- Reducing noise and interference
- Screening and Shielding

PID Control through SCADA

- The importance of timing and scan time
- When PID is not always appropriate: -Intermittent measurements

-Long transport delays
- Best methods of implementing PID Control for a SCADA system

Data Communications and Networking

- Data transfer from PLC/RTU, Report Generation
- Background to cables
- Noise and interference on cables
- Twisted pair cables and Fibre optic cables
- Public network provided services

Industrial Communications and Protocols

- RS-232 interface standard
- RS-485 interface standard
- Fieldbus; Modbus; DNP3.0

SCADA Network Security

- Introduction
- Authentication and encryption
- SCADA firewalls
- Firewall architectures and guidelines
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